



Key Points:

- Smoke has a substantial impact on particulate matter concentrations and health across the entire US
- We estimate that smoke caused $24\text{--}62 \times 10^3$ asthma emergency department visits for 2019–2024
- Smoke has a significant impact on regulatory compliance with the US annual $\text{PM}_{2.5}$ standard

Supporting Information:

Supporting Information may be found in the online version of this article.

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Health and Regulatory Impacts of $\text{PM}_{2.5}$ From Wildland Fires for 2019–2024 in the US

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Abstract In recent years the area of wildland fires has increased in the US and many regions have experienced extremely poor air quality due to smoke. We combine data for $\text{PM}_{2.5}$ (particulate matter with diameter $<2.5 \mu\text{m}$) with a satellite product to identify smoke-influenced days and associated $\text{PM}_{2.5}$ for 2019–2024 from air quality monitors covering 85% of the US population. Averaged across the US, smoke is present on 14.2% of all days and increases the daily mean $\text{PM}_{2.5}$ concentration on these days by $6.9 \mu\text{g m}^{-3}$, with a maximum of $687 \mu\text{g m}^{-3}$. For each region, we estimate the contribution of smoke $\text{PM}_{2.5}$ to emergency department visits (EDV) for asthma. While rural regions in the western US have the highest contributions of smoke to the $\text{PM}_{2.5}$ concentrations, larger metropolitan areas have a greater number of EDV from smoke due to greater populations. We next consider the impacts of smoke on compliance with the US annual standard for $\text{PM}_{2.5}$ ($9.0 \mu\text{g m}^{-3}$). Using data for 2022–2024, out of 807 of monitors studied 174 would not meet the standard, but in the absence of smoke, only 57 of these would not meet the standard. While observed annual $\text{PM}_{2.5}$ has shown no significant change over the past 6 years, we find a significant decline in $\text{PM}_{2.5}$ when the smoke contributions are excluded.

Plain Language Summary Impacts of wildfires in the US have increased substantially in the recent decade. We use surface observations to identify the smokiest regions in the US and estimate daily emergency departments for asthma due to smoke. While regions in the Western US have the highest overall impacts (in absolute amount of smoke), the impacts of smoke are felt across the US and in Alaska. Smoke also has a significant influence on compliance with the US air quality standards for $\text{PM}_{2.5}$.

1. Introduction

Wildfires are now a major source of emissions contributing to poor air quality in the US. In the past decade (2015–2024), wildland fires burned an average of 3.1 million hectares in the US, which is more than twice the amount in the years 1983–2000 (NIFC, 2025). In a period when industrial emissions are declining (US EPA, 2025a), emissions from wildland fires are now a major contributor to degraded air quality for both $\text{PM}_{2.5}$ (particulate matter less than $2.5 \mu\text{m}$ in diameter; Burke et al., 2023; McClure & Jaffe, 2018; O'Dell et al., 2019) and ozone (O_3) (Lee & Jaffe, 2024). In addition to US fires, wildland fires emanating from Canada and Mexico are major contributors to poor air quality throughout North America (Kaulfus et al., 2017). The year 2023 was exceptional for fires in Canada, with more than 15 million hectares burned (Jain et al., 2024; Kirchmeier-Young et al., 2024; Kolden et al., 2024); these fires resulted in substantial air quality impacts in both Canada (Jain et al., 2024) and the US (Lee & Jaffe, 2024). There are several causes for this increase in fires including climate change (Abatzoglou et al., 2019; Halofsky et al., 2020; Jain et al., 2024; Kirchmeier-Young et al., 2024; Williams et al., 2019) and historical forest management practices (e.g., Hagmann et al., 2021).

Wildland fires emit many different compounds including gases and particulate matter. Most health studies, and the regulatory standards, focus on $\text{PM}_{2.5}$. Other important compounds include nitrogen oxides, which contribute to O_3 formation (Jaffe & Wigder, 2012), and hundreds of volatile organic compounds (VOCs, Akagi et al., 2011). While many of these pollutants are likely associated with health effects, $\text{PM}_{2.5}$ is thought to be the most hazardous and a useful surrogate for the full suite of emissions (O'Dell et al., 2020). Numerous studies have documented health impacts from smoke on cardiovascular and respiratory hospitalizations (e.g., Borchers Arriagada

et al., 2019; Fann et al., 2018; Hahn et al., 2021; J. C. Liu et al., 2017; Reid et al., 2019) and asthma (Gould et al., 2024; Jiao et al., 2024; Noah et al., 2023). Other studies have also documented increased pre-mature mortality associated with smoke (e.g., Connolly et al., 2024; Doubleday et al., 2020; Gould et al., 2024; Ma et al., 2024). Smoke is thought to be more hazardous to respiratory health (Aguilera et al., 2021; Borchers Arriagada et al., 2019; Wegesser et al., 2009) and risk of dementia (Zhang et al., 2023) than other types of $PM_{2.5}$, and more aged smoke aerosols likely have greater health impacts, per unit of mass, than other types of aerosols due to greater aerosol oxidation (Bates et al., 2019; Holder et al., 2012; Joo et al., 2024; Wong et al., 2019). Elser et al. (2025) have demonstrated an 18% increased risk of dementia for each $1\text{-}\mu\text{g m}^{-3}$ increase in smoke $PM_{2.5}$, averaged over 3-years.

The meta analysis of relative risks (RR) by Borchers Arriagada et al. (2019) indicates greater health impacts from smoke $PM_{2.5}$ compared to other sources of $PM_{2.5}$. From their review, they calculate a relative risk (RR) of 1.07, with a 95% CI of 1.04–1.09 for asthma ER visits per $10\text{ }\mu\text{g m}^{-3}$ exposure to smoke (all populations). For non-smoke $PM_{2.5}$ exposure, we find RR values of 1.018 (95% CI of 1.005–1.028) from Lim et al. (2016), 1.015 (95% CI of 1.012–1.017) from Fan et al. (2016) and a mean of 1.02 from three studies reviewed by Anenberg et al. (2018). Thus it appears that $PM_{2.5}$ from wildfires consistently shows greater health impact than $PM_{2.5}$ from other sources. However, we acknowledge the significant uncertainty in each of the individual studies included in the meta analyses, as well as the compounded uncertainty when study results are synthesized. Given differences in exposure assessment between smoke and non-smoke $PM_{2.5}$ as well as potential underlying vulnerabilities of study populations and unmeasured confounders (e.g., behavior), there is likely to be a range of effects that can be observed. For scientific inference and policy discussions, it is important for our estimates to reflect that uncertainty.

Smoke from wildland fires can also have a significant regulatory impact for locations that fail to meet air quality standards. In the US the “exceptional events rule” (EER) allows states and tribal agencies the ability to request exclusions of air quality data on exceedance days if the source of the pollution was uncontrollable, for example, wildland fires (US EPA, 2016). However, the process is complex and time consuming for state agencies, and the exact criteria for exclusion of data are ambiguous. Thus, identifying specific criteria for smoke influence on air quality would be desirable for state and tribal agencies. Given the recent tightening of the $PM_{2.5}$ annual standard to $9\text{ }\mu\text{g m}^{-3}$ averaged over a 3-year period (US EPA, 2024b), identifying wildfire smoke influence at individual monitoring locations is important for state efforts to meet air quality standards. We note that the daily $PM_{2.5}$ standard was not tightened at the same time as the annual standard. Nonetheless, identifying wildfire smoke and removing it from regulatory implications does not reduce the potential health impacts from population exposure.

Various methods have been employed to identify wildfire smoke and corresponding health impacts at the surface, including satellite data, transport modeling and/or machine learning models (Childs et al., 2022; Considine et al., 2023; Gong et al., 2017; Lee & Jaffe, 2024; Qiu et al., 2024). One common tool is the NOAA Hazard Mapping System Fire and Smoke Product (hereafter simply HMS). The daily HMS smoke product is derived from multiple satellite products and identifies fire locations and smoke plume extent (Kaulfus et al., 2017; Ruminski et al., 2011). While satellite data is critical to identify fire locations, emissions and smoke transport, satellite data alone is not sufficient to identify air quality impacts at the surface without additional information (T. Liu et al., 2024). For example, Buysse et al. (2019) found that $PM_{2.5}$ was enhanced at the surface on an average of 30%–70% of days on days when there was overhead HMS smoke, for 18 Western US cities. While large $PM_{2.5}$ enhancements due to wildland fires, up to hundreds of $\mu\text{g m}^{-3}$, are relatively easy to identify, more modest increases are less obvious. Because of the complexity of most exposure assessment approaches, these methods can be difficult for state and local air quality managers to employ to show smoke influence in a clear and timely manner.

In this analysis, we use a satellite-derived smoke indicator, combined with surface $PM_{2.5}$ at state, federal and tribal air quality stations to identify smoke influenced $PM_{2.5}$ on every day for 2019–2024 at each site. This analysis includes 1,455 individual air quality monitors in all 50 states and the District of Columbia. To evaluate the health impacts of smoke on human populations, we group the data by Core Based Statistical Areas (CBSA) or counties, for monitors that are not included in a CBSA. In total, our analysis includes 358 US CBSAs plus 74 US counties with air monitors, covering more than 85% of the US population. Our results demonstrate the effectiveness of a simple method to estimate smoke impact at air quality monitors, which could be used by state/local agencies to rapidly estimate smoke impacts on $PM_{2.5}$.

2. Methods

Daily surface $PM_{2.5}$ data for January–December of 2019–2024 were obtained from the US EPA's Air Quality System (AQS) network (US EPA, 2024a). This includes data that were collected by federal, state, local and tribal agencies using consistent procedures and quality control. For the 2019–2023 data, all data were considered final and validated by the US EPA, whereas for 2024 data, 78% were final, but the remainder (22%) were only available as preliminary data and these 2024 data were included in our analysis. In past work, we have found that most of the preliminary data passes to final data without any changes. Details on the data collection, data selection and quality control are given in the SI.

To determine the presence of smoke plumes, we use the HMS smoke product which uses visible imagery from multiple satellites (>6) to estimate smoke extent on a daily basis (NOAA, 2025; Ruminski et al., 2011). The product characterizes each smoke plume by intensity (light, medium or heavy). While the HMS product has proven to be very useful in a number of studies on the distribution of smoke, it does have limitations, especially at night and in the presence of clouds. For each monitoring location, we use the HMS smoke product to identify the presence ($HMS = 1$) or absence ($HMS = 0$) of an overhead smoke plume. For our analysis a “smoke day” is identified when both surface $PM_{2.5}$ is elevated above a set criteria and there is an overhead smoke plume of any intensity ($HMS = 1$). We use two $PM_{2.5}$ smoke criteria representing “probable smoke day” (criteria 1) and “highly probable smoke day” (criteria 2). For criteria 1 we use $\frac{1}{2}$ the median absolute deviation (MAD) and for criteria 2 we use 1 MAD of the $HMS = 0$ data.

$$PM_{2.5} \text{ criteria 1} = \text{median } PM_{2.5} (HMS = 0 \text{ days}) + \frac{1}{2} \text{ MAD } (HMS = 0 \text{ days}) \quad (1)$$

$$PM_{2.5} \text{ criteria 2} = \text{median } PM_{2.5} (HMS = 0 \text{ days}) + \text{MAD } (HMS = 0 \text{ days}) \quad (2)$$

By definition, all days that meet criteria 1 will also meet criteria 2, but not vice versa. These criteria are calculated at each site on the $HMS = 0$ data by month, to account for geographic and seasonal variations. We use all monthly data across the 6-year period (e.g., January 2019, 2020, etc.) to calculate each $PM_{2.5}$ level. It is necessary to use multiple years, as some months have very few non-smoke days at some locations. While there can be modest trends in the non-smoke $PM_{2.5}$, at some locations, these are relatively small compared to the smoke impacts. More discussion on trends on the non-smoke days is provided in the results section.

Criteria 1 (probable smoke day) and 2 (highly probable smoke day) represent approximately the 67th and 80th percentiles of the $HMS = 0$ distribution. In our previous work, we used a similar procedure (Gong et al., 2017; Lee & Jaffe, 2024), but in those studies the $PM_{2.5}$ criteria was based on the mean and standard deviation of the $HMS = 0$ data. The MAD is a more appropriate statistical metric than the standard deviation for non-gaussian data sets (Leys et al., 2013), such as $PM_{2.5}$ data.

The highly probable smoke category includes slightly fewer smoke days compared to the probable smoke category since the $PM_{2.5}$ smoke criteria is greater. Using the highly probable classification we find that 11.3% are identified as smoke influenced, compared to 14.2% of all days for the probable classification.

The $PM_{2.5}$ due to smoke is calculated from:

$$PM_{2.5} (\text{smoke}) = PM_{2.5} (\text{observed}) - PM_{2.5} (\text{criteria}) \quad (3)$$

$$PM_{2.5} (\text{nonsmoke}) = PM_{2.5} (\text{observed}) - PM_{2.5} (\text{smoke}) \quad (4)$$

The $PM_{2.5}$ from smoke is only calculated on smoke days, when both $HMS = 1$ and $PM_{2.5}$ is greater than the $PM_{2.5}$ criteria, so this value is always positive. Figure S1 and S2 in Supporting Information S1 shows distributions of these criteria by month. On non-smoke days, $PM_{2.5} (\text{smoke}) = 0$ and therefore $PM_{2.5} (\text{non-smoke}) = PM_{2.5} (\text{observed})$.

To assess the health impacts, the individual EPA monitor data are aggregated each day by the Core Based Statistical Area (CBSA) or county, if a monitoring site is not located in a CBSA. The mean $PM_{2.5}$ value and $PM_{2.5}$ (smoke) from all monitors is assumed to apply to the entire population in that CBSA or county. For monitors that are not in a designated CBSA, we use the county name and population to evaluate health impacts.

We include all CBSAs and counties that have at least 80% daily data coverage for 2019–2024, which includes 358 US Core Based Statistical Areas plus 74 counties, covering 85% of the US population. For the 432 CBSAs plus counties, there is an average of 2.1 monitors per CBSA or county (range 1–15). While smoke has multiple health impacts, we focus on one metric: daily asthma emergency department visits (EDV). We focus on this metric as it shows up in almost every analysis as having a positive relationship with smoke exposure. Also, EDV captures a wider swath of the population: hospitalizations may only represent more severe cases, so EDV is a more sensitive indicator to examine the influence of smoke on health outcomes. The daily EDV due to smoke are calculated similar to the methods used by in O’Dell et al. (2021), with input data from AHRQ (2006) and Borchers Arriagada et al. (2019), but we apply the higher relative risk values only on the additional $PM_{2.5}$ due to smoke. Equations 5 and 6 show the calculation method

$$\text{Asthma EDV (nonsmoke)} = K * (1 - \exp^{(-\beta_{\text{nonsmoke}} * PM_{2.5}(\text{NS}))}) \quad (5)$$

$$\text{Asthma EDV (smoke)} = K * [\exp^{(-\beta_{\text{smoke}} * PM_{2.5}(\text{NS}))} - \exp^{(-\beta_{\text{smoke}} * PM_{2.5}(T))}] \quad (6)$$

where:

K = population * the annual baseline rate for asthma EDV per unit population/365

$\beta = \ln(RR)/10$

RR is the relative risk per $10 \mu\text{g m}^{-3}$ of $PM_{2.5}$.

$PM_{2.5}(\text{NS})$ and $PM_{2.5}(T)$ refer to the non-smoke $PM_{2.5}$ and total $PM_{2.5}$, respectively.

We use an annual baseline rate of 625.7 per 100,000 population and a relative risk of 1.07 per $10 \mu\text{g m}^{-3}$ of smoke and a relative risk of 1.02 per $10 \mu\text{g m}^{-3}$ for non-smoke $PM_{2.5}$.

In Supporting Information S1, we also show summary results for a range of RR values for smoke (1.04, 1.07, and 1.09) and for both probable and high probability smoke days. Due to the non-linearity in the relationship (Equation 5), we assess the smoke contribution to the daily EDV, by first calculating EDV for the observed $PM_{2.5}$, then calculate EDV for the non-smoke $PM_{2.5}$. The smoke contribution to the EDV is determined from the difference. By definition, the contribution of smoke to EDV on non-smoke days is zero.

To assess the regulatory impacts on the 2024 $PM_{2.5}$ annual and daily standards, we include all monitors with a valid design value for 2022–2024 (US EPA, 2025b) and that overlap our selected monitors. This period is the initial 3-year window during which compliance with the new annual $PM_{2.5}$ standard will be determined. In total, this includes 807 monitors in all 50 states and D.C.

There are several uncertainties in our identification of $PM_{2.5}$ due to smoke. This includes missing fires or smoke plumes in the HMS data (small fires, clouds, etc.), or small or fast-moving fires that do not impact a regulatory monitoring location. In addition, this method will not distinguish between wildfires, prescribed or agricultural burning. More details on the method and further discussion of uncertainties are given in the Supplemental Information, along with a comparison of our smoke $PM_{2.5}$ values with another recent study that published data for 2019–2022 (see Figure S3 in Supporting Information S1) and supports our use of the “probable” smoke day classification as the best indicator of smoke impacts.

Some of the key improvements in our method over previous work is the relatively simple algorithm, which means that smoke days can be identified quickly and by non-experts for each CBSA and individual monitor. However, we note that this does sacrifice higher spatial resolution afforded by more complex statistical and model-based approaches. Another important improvement is that our analysis includes all 50 states and D.C. Finally, we have developed a public R-Shiny app, where these results can be accessed by other scientists (e.g., for health studies) or policy makers (e.g., for regulatory implications) (<https://smokelyze.netlify.app>).

3. Results

Figure 1 shows the occurrence of smoke days by month and year for five regions of the country: Western US (continental US $> 102^\circ\text{W}$), Central US ($82^\circ\text{--}102^\circ\text{W}$), Eastern US ($< 82^\circ\text{W}$), Alaska, and Hawaii. Figure 1 also shows the average concentration of $PM_{2.5}$ on smoke days within each region (1c) and the trend in smoke and non-smoke $PM_{2.5}$ for 2019–2024. Smoke days are most common from May–October in the Central and Eastern US,

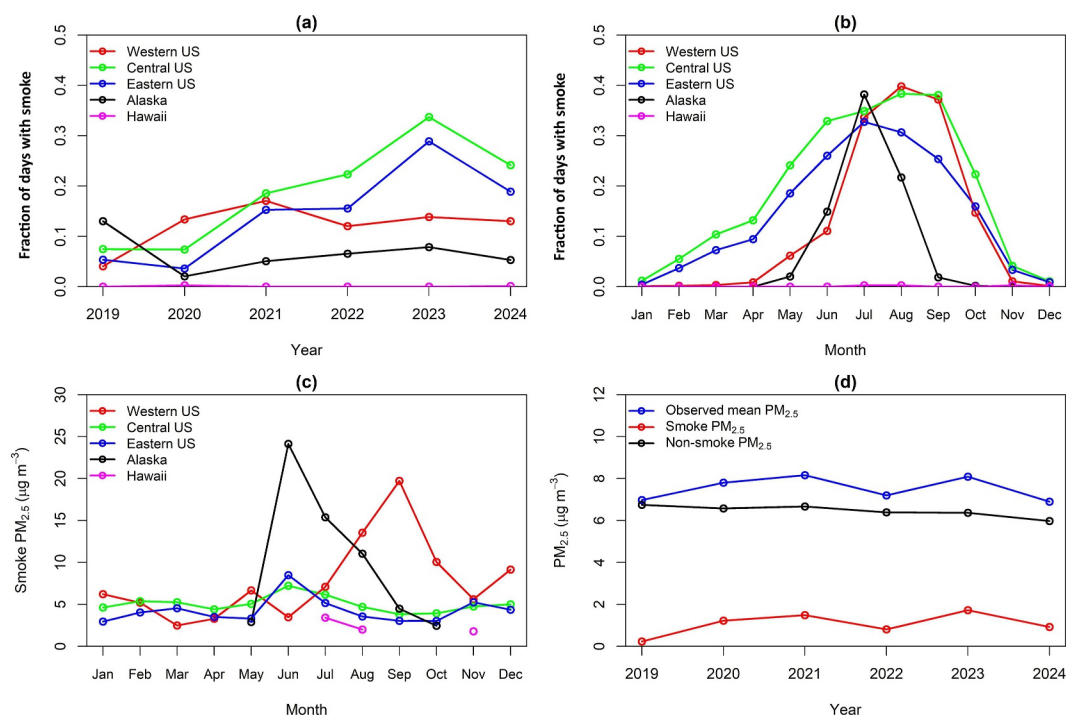


Figure 1. Fraction of days with smoke for each region by (a) year and (b) month. (c) Quantity of $PM_{2.5}$ added by smoke on smoke days. For Hawaii the fraction of smoke days is very low ($<5 \times 10^{-4}$). (d) Annual mean observed, smoke and non-smoke $PM_{2.5}$ for the entire US. The downward trend in non-smoke $PM_{2.5}$, shown in panel (d), is statistically significant ($p < 0.01$). Plots similar to panel (d) for individual regions are shown in Figure S5 in Supporting Information S1.

July–September in the Western US and July in Alaska. From our analysis, we identified only a small number of smoke days in Hawaii, despite occasional large fire outbreaks on the islands (e.g., August 2023). This can be explained by the relative paucity of $PM_{2.5}$ monitors in Hawaii and the fact that most smoke there will quickly be transported out over the ocean. In terms of years, 2023 had the greatest occurrence of smoke days due to the substantial influence from the huge Canadian wildfires that year. In the Western US, 2020 and 2021 had the greatest occurrence of smoke days (17% and 21%, respectively), although 2023 and 2024 were not far behind. Of the years studied, 2019 has the lowest smoke influence across most of the continental US. This reflects a low area of wildland fires burned in the continental US (0.9 million ha, NIFC (2025)). In contrast, Alaska had its highest smoke $PM_{2.5}$ in 2019.

The downward trend in $PM_{2.5}$ on non-smoke days (Figure 1d) is statistically significant, despite the relatively short time covered by this analysis. This indicates a significant decline in the non-smoke sources of $PM_{2.5}$, a change of $0.8 \mu\text{g m}^{-3}$ or 12% over this time frame. This decline is not present in the original $PM_{2.5}$ data and can only be detected if the smoke influence is removed. Figure S4 in Supporting Information S1 shows this relationship by region. In every region except for Hawaii, there is a downward trend in the non-smoke $PM_{2.5}$. We note that non-wildfire emissions of primary $PM_{2.5}$ and gaseous precursors (SO_2 , NO_x , VOCs and NH_3) decreased by 35% during this period (U.S. EPA, 2026). While we did not account for a trend in identifying smoke days, we do not feel this has a strong influence on our analysis given the magnitude of the trend, compared to the amount of smoke on smoke days.

Figure S5 in Supporting Information S1 shows histograms of $PM_{2.5}$ on smoke and non-smoke days and the amount of $PM_{2.5}$ added on smoke days. On smoke days, the mean observed $PM_{2.5}$ is $13.7 \mu\text{g m}^{-3}$ and we calculate that the mean enhancement due to smoke on these days is $6.9 \mu\text{g m}^{-3}$. Figure 1c shows the $PM_{2.5}$ contributions by region. The largest mean $PM_{2.5}$ enhancements are found in Alaska ($15.2 \mu\text{g m}^{-3}$) and the Western US ($12.0 \mu\text{g m}^{-3}$), with smaller amounts for the Central US ($5.14 \mu\text{g m}^{-3}$), Eastern US ($4.53 \mu\text{g m}^{-3}$) and Hawaii ($2.4 \mu\text{g m}^{-3}$). This likely reflects the fact that smoke in the Western US or Alaska is from nearby fires, whereas smoke in the Central or Eastern US has likely been transported a greater distance and experienced more dilution.

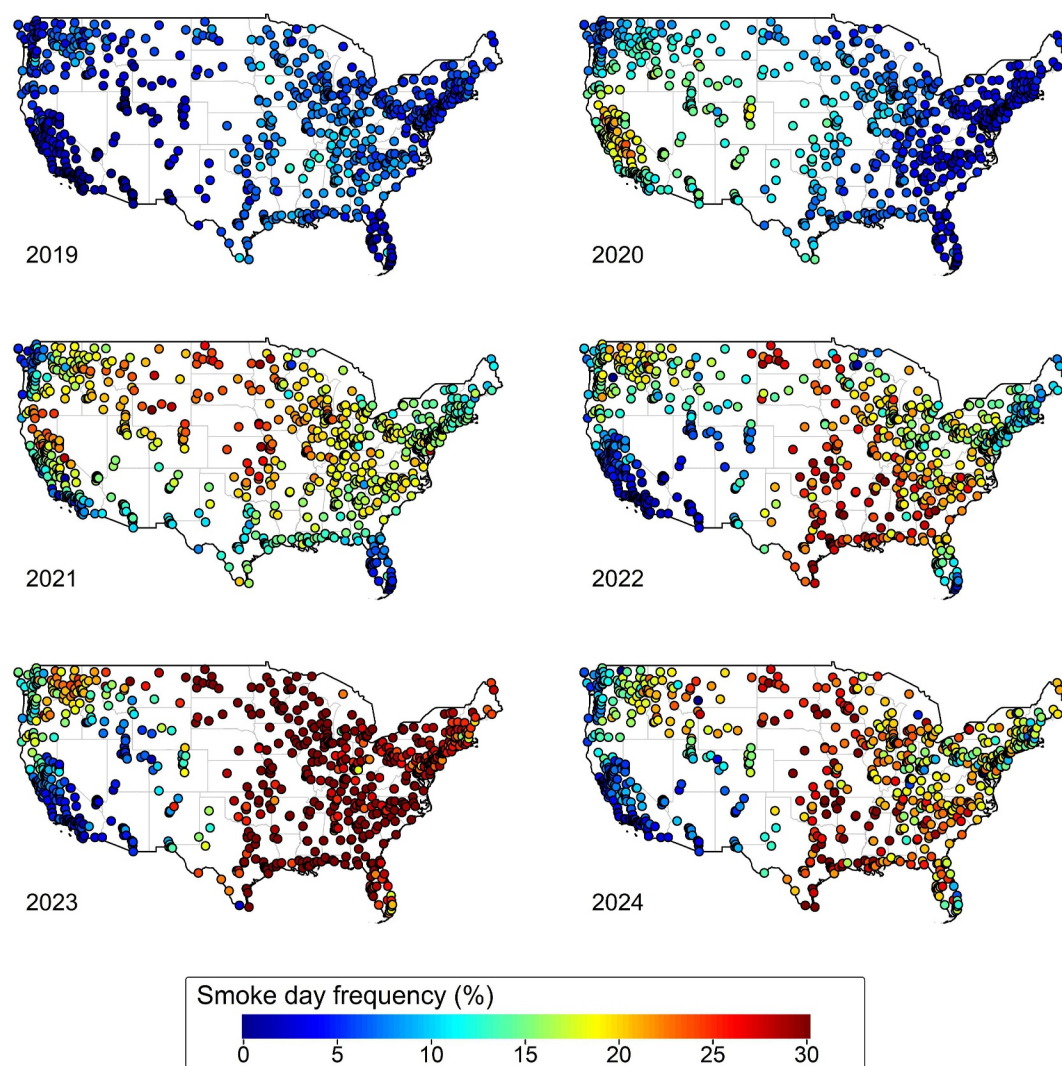


Figure 2. Fraction of smoke days across the continental US for 2019–2024 based on individual monitor values.

Figure 2 shows more details on the distribution of smoke days by year. In 2020 and 2021, California had exceptional wildfire years with 1.7 and 1.1 million hectares burned, respectively (NIFC, 2025). For all of California, the wildfire smoke day fraction for the full year was 0.17 and 0.15 in 2020 and 2021, respectively. While 2020 also had an elevated smoke day frequency throughout the western US (10%–15%), 2021 was even higher at 15%–20%. In the Central US nearly every year has significant smoke day occurrences, but 2023 was by far the greatest, due to transported smoke from Canada. A similar pattern is seen in the Eastern US, although the smoke day frequency is not as high as in the Central US.

Figure 3 shows the calculated smoke $PM_{2.5}$ for every CBSA and county in our data set. This is shown as the mean for all days in 2019–2024. Since smoke only occurs on a fraction of days, the actual $PM_{2.5}$ on smoke days is much higher than these values.

Most parts of the Pacific Northwest and northern California have a mean smoke $PM_{2.5} > 1 \mu g m^{-3}$, averaged over every day of 2019–2024. One heavily impacted region (Trinity County, CA) has more than $7 \mu g m^{-3}$ of smoke and many CBSA in the western US have more than $2 \mu g m^{-3}$ over this period. All of these sites in the Western US had extremely high monthly mean values in either 2020 or 2021. As one example, Trinity County, CA had a monthly mean smoke $PM_{2.5}$ of $249 \mu g m^{-3}$ in August of 2021, along with high values at many other western sites in that same month. Mono and Mariposa counties, CA had monthly means over $100 \mu g m^{-3}$ in 2020, along with Salem and Albany, OR. Other high monthly means include several sites in Oregon in August 2023

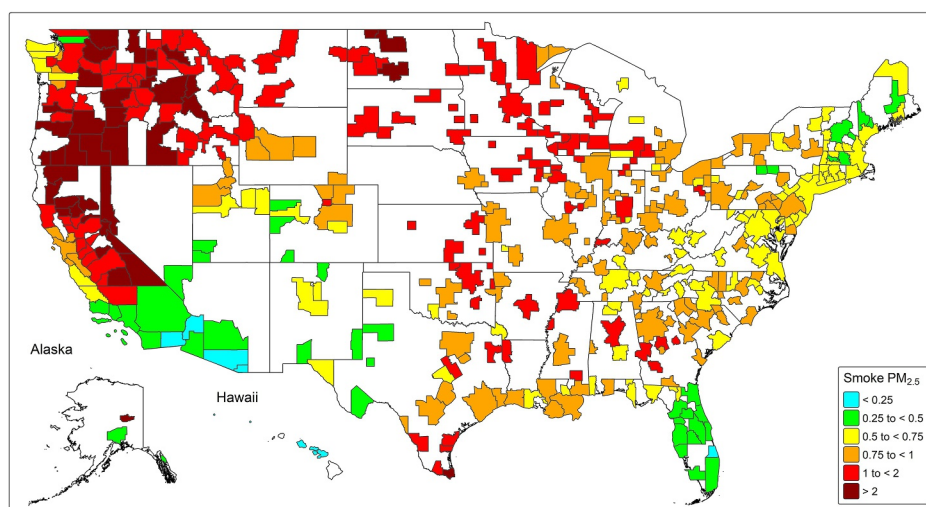


Figure 3. Mean smoke $\text{PM}_{2.5}$ ($\mu\text{g m}^{-3}$) for all CBSAs and counties in our analysis for 2019–2024. This is shown as the mean for all days in 2019–2024. Since smoke only occurs on a fraction of days, the actual $\text{PM}_{2.5}$ on smoke days is much higher than these values.

(33–53 $\mu\text{g m}^{-3}$), sites in Wisconsin in June 2023 (26–27 $\mu\text{g m}^{-3}$) and sites in Fairbanks, AK in July 2019 (39 $\mu\text{g m}^{-3}$).

Table 1 shows the top 25 CBSAs or counties in terms of smoke $\text{PM}_{2.5}$ and the fraction of observed $\text{PM}_{2.5}$ that is due to smoke. For these heavily impacted regions, smoke makes up 20%–54% of the total $\text{PM}_{2.5}$ concentrations. Some of the highest mean smoke $\text{PM}_{2.5}$ values occurred in 2020 across the Western US, when California, Oregon and Washington had population weighted smoke concentrations of 2.7, 6.4, and 3.0 $\mu\text{g m}^{-3}$ averaged over the entire year (calculated by weighting the daily statewide mean $\text{PM}_{2.5}$, by the population in each CBSA or county). These values are similar to the population weighted smoke $\text{PM}_{2.5}$ values reported by Childs et al. (2024) for the same time period for California (2.9 $\mu\text{g m}^{-3}$) and Oregon (6.4 $\mu\text{g m}^{-3}$). In 2023, the worst smoke occurrences moved to the Central and Eastern US. For example, Minnesota, Michigan, and New York had population weighted smoke $\text{PM}_{2.5}$ values in 2023 of 3.3, 3.0, and 2.3 $\mu\text{g m}^{-3}$, respectively.

Table 1 also shows the cumulative asthma EDV for the top 25 “smokiest” regions for 2019–2024 and Table 2 shows the same information for the regions with the highest smoke-attributable asthma EDVs. For the entire US, we estimate 5.1×10^4 asthma EDVs over the 6-year time period (2019–2024) or an average of 8.5×10^3 per year. Table S3 in Supporting Information S1 shows the results for EDVs with different assumptions of RRs and for the probable versus highly probable smoke day categories. The range in smoke EDVs is $2.4\text{--}6.2 \times 10^4$, where the uncertainty in the smoke RR values makes the largest contribution to this range. For non-smoke $\text{PM}_{2.5}$ we estimate $1.51\text{--}1.55 \times 10^5$ cumulative asthma EDVs over this time period. Thus we estimate that smoke is responsible for 34% (range of 16%–41%) of the non-smoke total EDV for this same time period. However, 2023 alone accounted for 33% of these, due to the high smoke $\text{PM}_{2.5}$ concentrations across the more populated regions of the country. Comparing Tables 1 and 2, we see that although rural parts of the western US tend to have the highest smoke $\text{PM}_{2.5}$ values and fraction of $\text{PM}_{2.5}$ that is due to smoke, the more densely populated CBSAs of the Eastern and Central US have greater absolute numbers of smoke-attributable asthma EDVs due to the much greater populations, in alignment with the results from O’Dell et al. (2021).

Figure 4 shows the portion of the US population that is exposed to different annual-average levels of smoke $\text{PM}_{2.5}$ in different years. The figure is bounded by 2019, a very low smoke year, and 2023, a high smoke year (see Figures 1 and 2). In 2019, only a small portion (0.2%) of the population, all in Alaska, was exposed to an annual mean of 1 $\mu\text{g m}^{-3}$ or more of smoke $\text{PM}_{2.5}$, the level cited by Elser et al. (2025) for an 18% increased risk of dementia. In contrast, in 2023, more than 200 million people (69.7% of the US population) were exposed to annual mean smoke levels of 1 $\mu\text{g m}^{-3}$ or more. Averaged over these 6 years, 49 million people were exposed to annual-mean smoke $\text{PM}_{2.5}$ levels of 1 $\mu\text{g m}^{-3}$ or more and 6 million were exposed to 2 $\mu\text{g m}^{-3}$ or more.

Table 1
Top 25 CBSAs or Counties Based on the Highest Smoke PM_{2.5} in the US For 2019–2024

Rank	CBSA or county	Population (k)	PM _{2.5} due to smoke (μg m ⁻³)	Fraction of PM _{2.5} due to smoke	EDV
1	Trinity County, CA	14.4	7.09	0.54	13
2	Mono County, CA	15.9	4.49	0.52	10
3	Siskiyou County, CA	45.6	4.10	0.45	30
4	Mariposa County, CA	19.8	3.74	0.35	11
5	Plumas County, CA	19.1	3.73	0.29	13
6	Medford, OR	228.8	3.71	0.41	154
7	Grants Pass, OR	89.4	3.55	0.37	55
8	Bend, OR	263.4	3.28	0.38	148
9	Klamath Falls, OR	69.2	3.19	0.30	45
10	Gardnerville Ranchos, NV-CA	56.4	2.87	0.38	30
11	Lemhi County, ID	8.1	2.77	0.26	5
12	Okanogan County, WA	44.3	2.71	0.29	23
13	Carson City, NV	57.0	2.61	0.37	27
14	Ravalli County, MT	44.9	2.50	0.36	23
15	Harney County, OR	7.2	2.47	0.22	4
16	Red Bluff, CA	72.2	2.41	0.30	34
17	Eugene-Springfield, OR	383.9	2.36	0.29	150
18	Bishop, CA	20.3	2.35	0.31	9
19	Yakima, WA	265.7	2.32	0.24	113
20	Fairbanks-College, AK	114.9	2.25	0.21	49
21	Burke County, ND	1.8	2.25	0.34	1
22	Lewiston, ID-WA	62.2	2.21	0.27	28
23	Salem, OR	445.0	2.19	0.27	117
24	Visalia, CA	536.6	2.19	0.17	233
25	Reno, NV	605.3	2.18	0.29	259

Note. Table also shows the population, fraction of PM_{2.5} due to smoke and cumulative EDV for each area. The full list of all CBSAs and counties in our analysis is given in Table S1 in Supporting Information S1.

Figure 5 shows the implications of smoke on compliance with the US EPA's new annual PM_{2.5} standard. To meet this standard a monitor must have an annual average PM_{2.5} concentration of 9.0 μg m⁻³ or less, averaged over 3-years. The initial evaluation period will be based on data for 2022–2024. Our analysis includes 807 sites. Of these, 633 will likely meet the new annual standard, but 174 (21.6%) will not. For those sites that would not meet the standard, we recalculate the annual averages excluding smoke PM_{2.5} and find that the majority (117) of these 174 sites, could meet the annual standard if all smoke PM_{2.5} were excluded. However, states would need to exclude up to several hundred days of daily PM_{2.5} smoke day values at each site to meet the standard. In this way, smoke will cause many sites to exceed the new PM_{2.5} standard.

The impacts of smoke on compliance with the daily PM_{2.5} standard are also important, though not as pronounced as the annual standard, given that smoke contributes an average of 5.8 μg m⁻³ on smoky days (see Figure 1c). Considering 768 monitors, we find that 33 sites have more than 15 days per year (averaged for 2022–2024) over 35 μg m⁻³, due to smoke. The US daily PM_{2.5} standard is 35 μg m⁻³, expressed as the annual 98th percentile of the daily values, averaged over 3 years. This is called the design value by the US EPA. For the US as a whole, we find that out of the 768 monitors that we considered only 35 have design values for the daily PM_{2.5} standard that are over the standard, but only 15 would if smoke days were excluded. The most smoke impacted regions with respect to meeting the daily PM_{2.5} standard are found in Oregon, North Dakota and Washington. This is consistent with our finding that out of nearly 15,000 smoke days across all monitors, only 2.4% of these days have a PM_{2.5} value that exceeds 35 μg m⁻³ (see Figure S5 in Supporting Information S1).

Table 2
Top 25 CBSAs or Counties Based on the Highest Cumulative Emergency Department Visits for 2019–2024

Rank	CBSA or county	Population (k)	PM _{2.5} due to smoke ($\mu\text{g m}^{-3}$)	Fraction of PM _{2.5} due to smoke	EDV
1	New York-Newark-Jersey City, NY-NJ	19417.2	0.72	0.10	3146
2	Chicago-Naperville-Elgin, IL-IN	9837.7	0.92	0.10	2048
3	Houston-Pasadena-The Woodlands, TX	7593.3	0.92	0.09	1628
4	Dallas-Fort Worth-Arlington, TX	8038.1	0.80	0.09	1499
5	Atlanta-Sandy Springs-Roswell, GA	6756.0	0.86	0.10	1358
6	Los Angeles-Long Beach-Anaheim, CA	13324.0	0.43	0.04	1281
7	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	6292.6	0.81	0.10	1113
8	Minneapolis-St. Paul-Bloomington, MN-WI	3727.0	1.23	0.16	1055
9	Washington-Arlington-Alexandria, DC-VA-MD-WV	6600.2	0.68	0.09	1011
10	Sacramento-Roseville-Folsom, CA	2599.3	1.72	0.21	945
11	Detroit-Warren-Dearborn, MI	4176.5	0.96	0.10	894
12	San Francisco-Oakland-Fremont, CA	4546.4	0.85	0.11	827
13	Seattle-Tacoma-Bellevue, WA	3897.1	1.03	0.16	816
14	Denver-Aurora-Centennial, CO	3015.9	0.90	0.12	625
15	St. Louis, MO-IL	2944.0	0.88	0.10	603
16	Boston-Cambridge-Newton, MA-NH	4731.1	0.51	0.08	566
17	San Antonio-New Braunfels, TX	2690.4	0.89	0.10	554
18	Portland-Vancouver-Hillsboro, OR-WA	2572.8	1.31	0.20	551
19	Indianapolis-Carmel-Greenwood, IN	2169.9	1.06	0.11	514
20	Austin-Round Rock-San Marcos, TX	2393.3	0.88	0.10	488
21	Riverside-San Bernardino-Ontario, CA	5517.4	0.39	0.04	486
22	Charlotte-Concord-Gastonia, NC-SC	2924.8	0.69	0.09	475
23	Kansas City, MO-KS	2241.3	0.90	0.12	475
24	McAllen-Edinburg-Mission, TX	1086.5	1.94	0.18	469
25	Pittsburgh, PA	2372.1	0.86	0.10	457

Note. Table also shows the population and the amount and fraction of PM_{2.5} due to smoke for each area. The full list of all CBSAs and counties in our analysis is given in Table S1 in Supporting Information S1.

4. Discussion and Conclusions

Wildland fires are now a major cause of poor air quality in the US and the area of wildland fire burned in the US has approximately doubled in the past decade (NIFC, 2025). In addition to US fires, smoke from Canadian and Mexican fires can also make significant contributions to surface PM_{2.5} in the US (Kaulfus et al., 2017; Kirchmeier-Young et al., 2024). Many regions of the country frequently experience very poor air quality due to smoke. Although all regions of the US experience smoke, the strongest impacts on air quality are seen in the northwest (especially Northern California and Oregon) and north-central regions. In some rural regions of the western US smoke contributes more than 50% of the total PM_{2.5}. While observed PM_{2.5} has shown no significant trend in the past 6 years for the US as a whole, separating out the influence of smoke we can identify a significant downward trend in PM_{2.5} on non-smoke days. This indicates that the industrial sources of PM_{2.5} have declined, but this reduction has been obscured by the increase in smoke PM_{2.5}.

Using previously published risk assessments, we estimate an average annual asthma EDVs for 2019–2024 due to smoke of 8.5×10^3 , with significant interannual variations. 2023 had the greatest number of annual asthma EDVs due to substantial smoke impacts over the more populated central and eastern US. In contrast O’Dell et al. (2021) estimated $\sim 3.0 \times 10^3$ asthma EDVs per year due to smoke for the period of 2006–2018. However, this period had much less smoke than the more recent years included in the present study. O’Dell et al. (2021) estimated a nationwide population weighted PM_{2.5} from smoke of $\sim 0.3 \mu\text{g m}^{-3}$ over this time period, whereas we find a value of

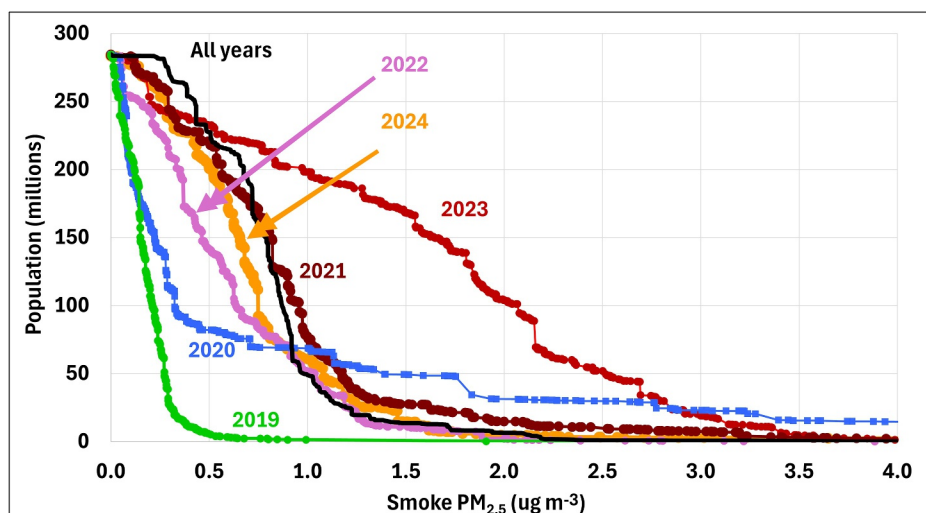


Figure 4. Population exposed to smoke $PM_{2.5}$ by year. Each line shows the number (millions) of US residents exposed to different amounts of smoke $PM_{2.5}$ each year. Note that the exposed regions change each year. This analysis covers 285 million people or 85% of the US population.

$0.81 \mu g m^{-3}$ for the same value for 2019–2024 (factor of 2.7 increase). Thus, our much higher annual estimate for asthma EDVs seems reasonable in this light.

There are several limitations to our work. First, a smoke day may be missed if the HMS smoke plume is not coincident with the monitor location. This can occur for several reasons, but most importantly is the presence of clouds, which will obscure the satellite's view of smoke plumes. An important caveat for all health studies based on ambient (outside) $PM_{2.5}$ concentrations is that we don't know how residents are responding to smoke. For example, some residents in smoke impacted areas may leave the area, while others remain. Or residents may remain, but in indoor locations with good air filtration. Thus, the ambient $PM_{2.5}$ measurements will not reflect the true human exposure for the entire population. At present, we have little quantitative data to explore these relationships. An additional limitation is that we assume exposure is equal across large counties and CBSAs. While

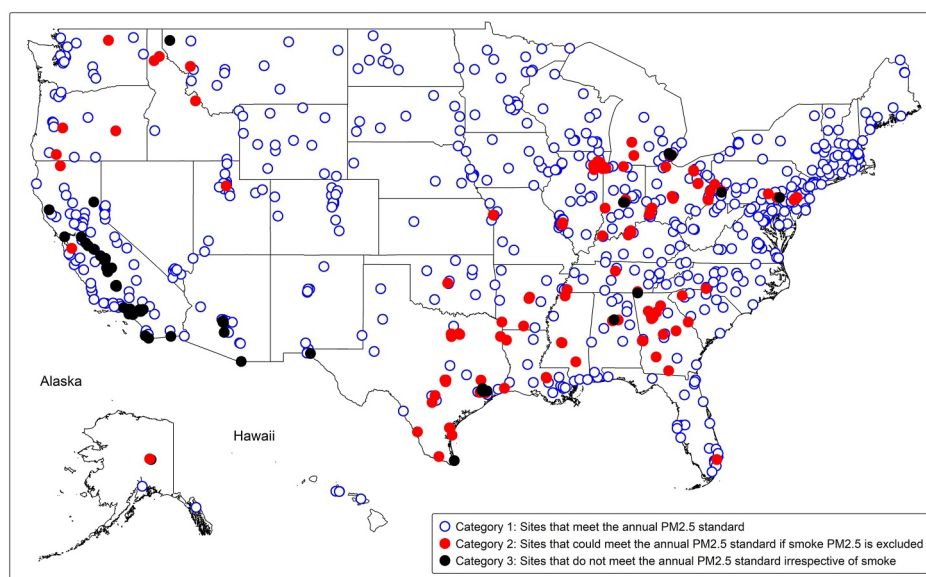


Figure 5. Impact of smoke on regulatory compliance with the annual $PM_{2.5}$ standard: Category 1: Sites with 2022–2024 average $PM_{2.5} < 9.0 \mu g m^{-3}$ ($n = 633$); Category 2: Sites with 2022–2024 average $PM_{2.5} < 9.0 \mu g m^{-3}$ only if smoke $PM_{2.5}$ is excluded ($n = 117$); Category 3: Sites with 2022–2024 average $PM_{2.5} > 9.0 \mu g m^{-3}$, regardless of smoke condition ($n = 57$).

this is in alignment with the levels at which air quality is regulated in the US, it does not follow patterns of smoke exposure which can vary greatly over small spatial scales. An additional limitation is the degree to which other pollutants, especially O_3 , are also enhanced with smoke along with higher temperatures (Lee & Jaffe, 2025). Finally, we assume that the RR used are uniform for smoke and non-smoke $PM_{2.5}$, whereas these likely vary across regions, age-groups and possibly economic status.

Knowing the pattern of smoke $PM_{2.5}$ in the US can help states and communities with emergency planning for high pollution days. Across the 6 years considered here, in absolute numbers the largest asthma EDV are found for the Central US (45%), followed by the Eastern (28%) and Western US (26%). In terms of individual states, the largest absolute numbers are seen in California, Texas, New York, Illinois, Pennsylvania, Georgia, Florida, Washington, North Carolina and Michigan. This reinforces a point first made in O'Dell et al. (2021), that the health impacts from smoke are not confined to the Western US. For all smoke impacted communities it is important to continue to, and increase, preparations for smoke impacts. This should include ensuring that all residents have access to information on air quality and air purifiers. In addition, these communities need to be prepared for increased respiratory disease and continue work to reduce uncontrolled fires, wherever possible.

Given the current frequency of smoke events across the country (Figure 2) and the impacts from smoke on meeting the annual $PM_{2.5}$ (Figure 5), we argue the EER process is not well suited for the present-day fire situation in the US. The US Clean Air Act allows states and tribal agencies to apply for data exclusion under the Exceptional Events Rule (EER). However, the process is complex and time-consuming for these agencies, and the criteria for a decision by the US EPA, is not clearly stated. The EER was developed to handle events such as stratospheric intrusions, impacting O_3 concentrations, dust storms and/or wildland fires, impacting both $PM_{2.5}$ and PM_{10} . However, the rule was developed at a time when wildfire events were much less frequent and the air quality standards were higher. Today, with the growing impacts of wildland fires on air quality and public health, merely excluding them as exceptional events is no longer viable. Removing the impact of wildfire smoke from the regulatory record does not remove the negative health impacts of smoke exposure on the population and disincentivizes state and local agencies to take action to reduce their population's smoke exposure. In order to maintain the original purpose of the National Ambient Air Quality Standards, "to protect public health and public welfare from harmful pollution", this process should be reevaluated.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Final data sets for this research can be found at the University of Washington's ResearchWorks archive (<https://hdl.handle.net/1773/53059>). A simplified data set can be explored in our user friendly app (<http://smokelyze.netlify>). This tool provides a data-driven tool for policymakers and researchers to assess wildfire smoke impacts on air quality at individual monitors, across states or for the whole US.

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